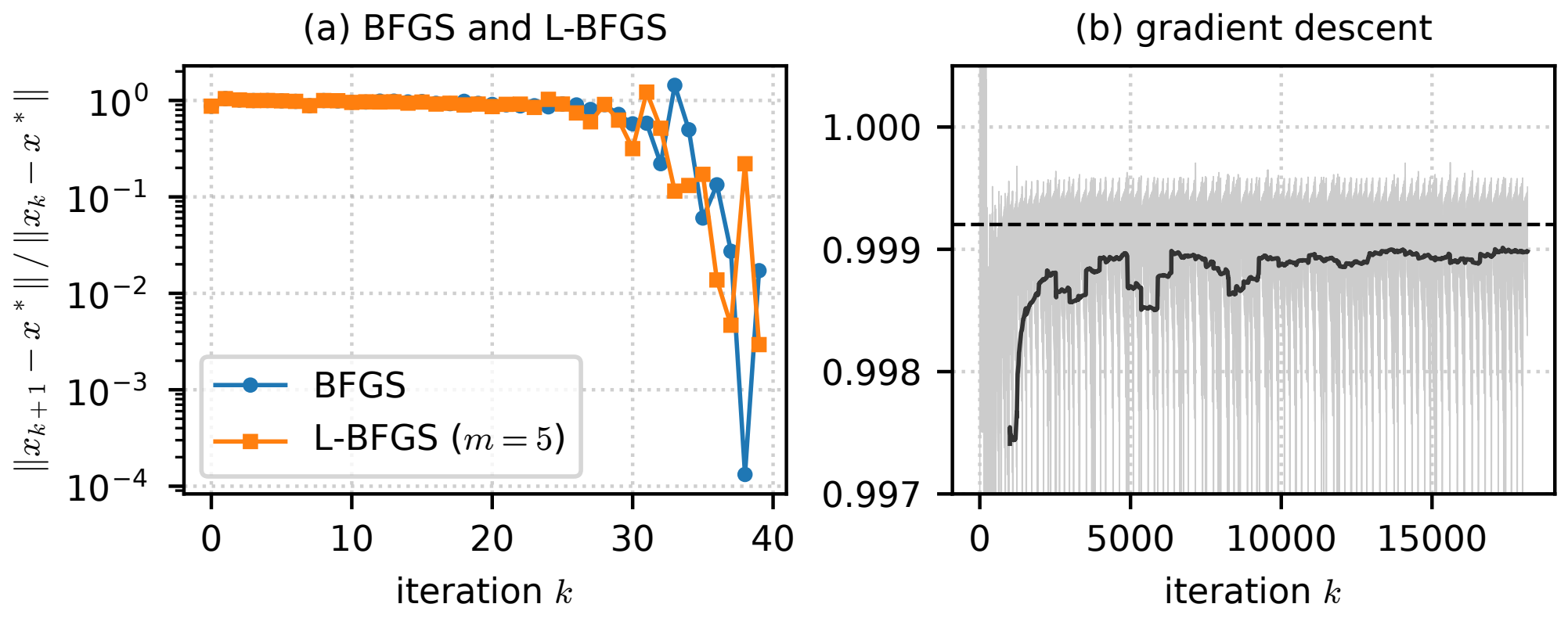

(a) BFGS and L-BFGS
(b) gradient descent
$\|x_{k+1} - x^*\| / \|x_k - x^*\|$
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
BFGS
L-BFGS ($m = 5$)
0
10
20
30
40
iteration k
1.000
0.999
0.998
0.997
0
5000
10000
15000
iteration k